WTO Accession, PNTR, and U.S. Import Quantities and Prices

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Outline

- Background and research question
- Data and empirical strategy
- Current results and robustness
- Next steps

Background

- China's economy grew rapidly in the 1990s, becoming a major exporter to the U.S.
- Under U.S. law, China received low tariff rates (Normal Trade Relations) but Congress had to **vote to renew** NTR status every year.
- This annual renewal created **trade-policy uncertainty**
  - importers faced the risk that tariffs could spike to 1930s-era levels if Congress voted no.
- In **October 2000**, the U.S. passed the law granting China Permanent Normal Trade Relations (PNTR), conditional on WTO accession.
- In **December 2001**, China officially joined the WTO; permanent low tariffs took effect in January 2002.
- We take advantage of the fact that the actual tariff rates mostly **did not change**: what changed was the **elimination of uncertainty** about future tariffs.

Research question

How did the resolution of U.S. trade-policy uncertainty around China starting in 2002 affect:

- the volume of imports from China into the U.S., and
- the prices U.S. buyers paid for Chinese imports?

Policy Timing and Identification Intuition

- **Pre-2001**: China receives NTR annually; risk of non-renewal creates uncertainty that is priced into trade decisions.
- **2000**: U.S. grants PNTR (legal permanence, conditional on WTO accession).
- **Dec 2001 / Jan 2002**: China joins WTO; permanent low tariffs take effect.

Why did uncertainty matter differently across products?

- Every product had two legislated tariff rates: a low NTR rate and a high non-NTR rate (dating back to the Smoot-Hawley Act of 1930).
- The **NTR gap** = non-NTR rate − NTR rate measures how much a product's tariff *would have jumped* if Congress didn't renew.
- Products with large gaps faced more risk → importers could be heavily impacted.
- Products with small gaps faced less risk → less impact by the annual vote.

Identification Strategy: Difference-in-Differences

Core idea

Compare import growth for **high-gap** products (more exposed to uncertainty) vs. **low-gap** products (less exposed), before and after WTO entry.

Why this works:

- Low-gap products serve as a **control group**: because, while they were still affected, they had a small stake from the policy change.
- Any "extra" import growth in high-gap products after 2002, beyond what low-gap products experienced, can be attributed to the removal of tariff uncertainty.

Treatment exposure (measured in 1999, before WTO negotiations):

$$\text{Exposure}_p = \text{gap\_pre}_p = (\text{non-NTR tariff})_p - (\text{NTR tariff})_p$$

Products above the median gap → "high-gap" (treatment); below → "low-gap" (control).

Data

Data sources

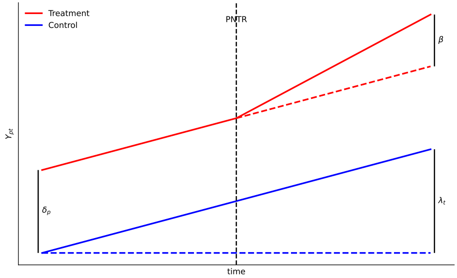
- USITC monthly imports by product and HTS.
- Tariff schedule information used to construct product-level `gap_pre`.

Baseline Specification

Two-way FE DiD

$$Y_{pt} = \beta\left(\text{HighGap}_p \times \text{Post}_t\right) + \delta_p + \lambda_t + \varepsilon_{pt}$$

- Y_{pt}: log import value, log quantity, or log unit value.
- δ_p: product fixed effects.
- λ_t: time fixed effects.
- ε_{pt}: error term.



DiD Ladder Regression Tables

ln_customs_value

Spec	$\hat{\beta}$	s.e.
m1	0.1109**	(0.0404)
m2	0.1251***	(0.0306)
m3	0.1124**	(0.0406)
m4	0.1211***	(0.0308)

ln_first_unit_qty

Spec	$\hat{\beta}$	s.e.
m1	0.2037***	(0.0534)
m2	0.1498***	(0.0358)
m3	0.2041***	(0.0536)
m4	0.1447***	(0.0361)

ln_unit_value

Spec	$\hat{\beta}$	s.e.
m1	-0.0926**	(0.0347)
m2	-0.0251	(0.0165)
m3	-0.0916**	(0.0347)
m4	-0.0239	(0.0165)

Specification ladder:

m1: no FE

m2: product FE

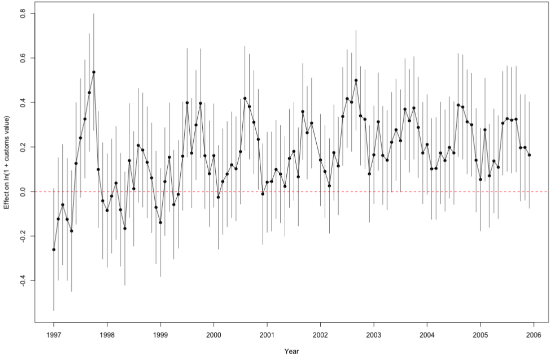
m3: date FE

m4: product + date FE

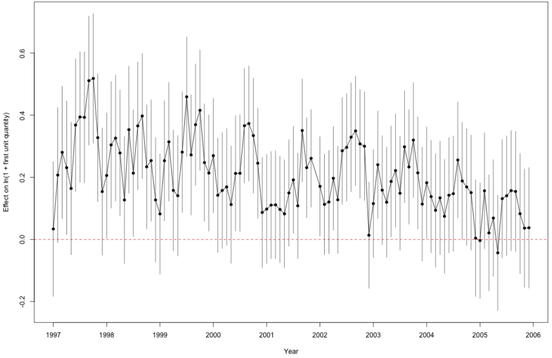
Event Study: Value and Quantity

Event-study specification

$$Y_{pt} = \sum_{k \neq -1} \beta_k \left(\text{HighGap}_p \times \mathbf{1}[\text{event\_time}_{pt} = k]\right) + \delta_p + \lambda_t + \varepsilon_{pt}$$



Value event study



Quantity event study

Takeaways and Next Steps

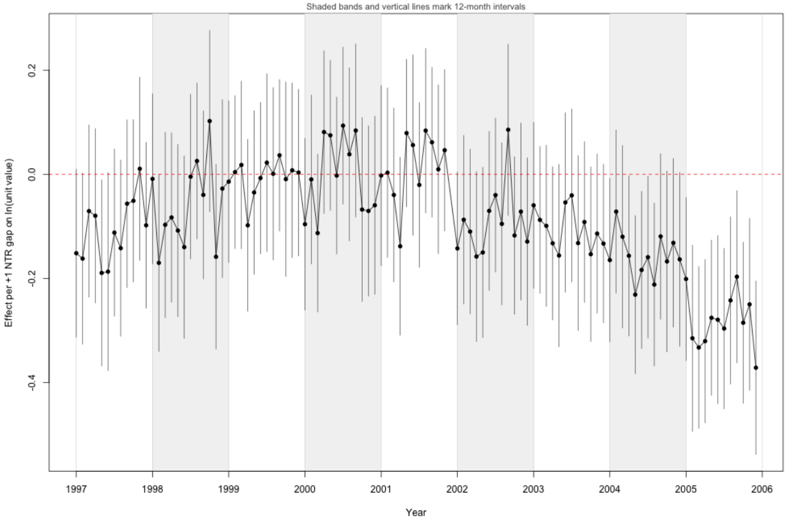
Takeaways

- A 10% increase in exposure causes an estimated $100(\exp(\hat{\beta} \cdot 0.10) - 1) \approx 1.3\%$ post-policy decline in "price" (unit value).
  - At the median exposure ($gap\_pre = 0.311$), the implied post-policy decline is $100(\exp(\hat{\beta} \cdot 0.311) - 1) \approx 4.0\%$.
  - Interpreted as the effect of removing trade-policy uncertainty on prices

Next steps

- Further finetune this parameter by:
  - seasonality controls
  - solve heterogeneity of products issue
  - controls for other events that occured
  - compare how imports changed for another country in their trade with China

Appendix: Seasonality



Questions?